Threat Intelligence Platform — UI Flow & Design Documentation

DISRUPTIONMATRIX · NAMMA AIR · PROOF-OF-CONCEPT BUILD · SYNTHETIC DATA

1. What this is

The platform is two linked products sharing one external-intelligence feed and one visual system:

- **DisruptionMatrix** — a global external-threat board. It correlates public-source events (weather, industrial action, geopolitical incidents, airport/airspace restrictions, regulatory change) and scores each on an **External Disruption Potential Score (EDPS, 0–100)**. It carries no airline data and never decides — it advises.
- **Namma Air Operations Threat Radar** — the same feed fused with Namma Air's own schedule, tails, crew pairings and station dependencies, producing an airline-specific **Namma Threat Score** that shows exactly how each external event exposes the BLR hub and DEL network.

A shared **Home** page lets a user pick a lens; every downstream page can jump back to Home or across to its counterpart.

2. Site map & navigation flow

Home — role/lens picker `(Home.dc.html)`

↓ choose a lens

DisruptionMatrix — global threat board `(DisruptionMatrix.dc.html)`

↓ world threat map

Global Threat Map — Leaflet map of all events `(Threat Map.html)`

↓ open threat detail

Event tabs: Assessment · Escalation & response · Event-type SOP · Retrospective

Namma Air Threat Radar — airline exposure board `(Namma Air Threat Radar.dc.html)`

↓ network map / data explorer

Namma Air Network Map — routes by threat `(Namma Air Threat`

Map.html)

Namma Air Data Explorer — schedule, fleet, crew, Gantt + solver (Namma Air Data Explorer.dc.html)

↑ "Fix schedule →" deep-links from the Radar's affected-rotations panel straight into the Gantt, pre-filtered to that threat

Every page carries: a link back to **Home**, a link to its counterpart product, and a persistent **light/dark theme toggle** — the choice is stored once (`localStorage: dm-app-theme`) and applies across all six pages immediately.

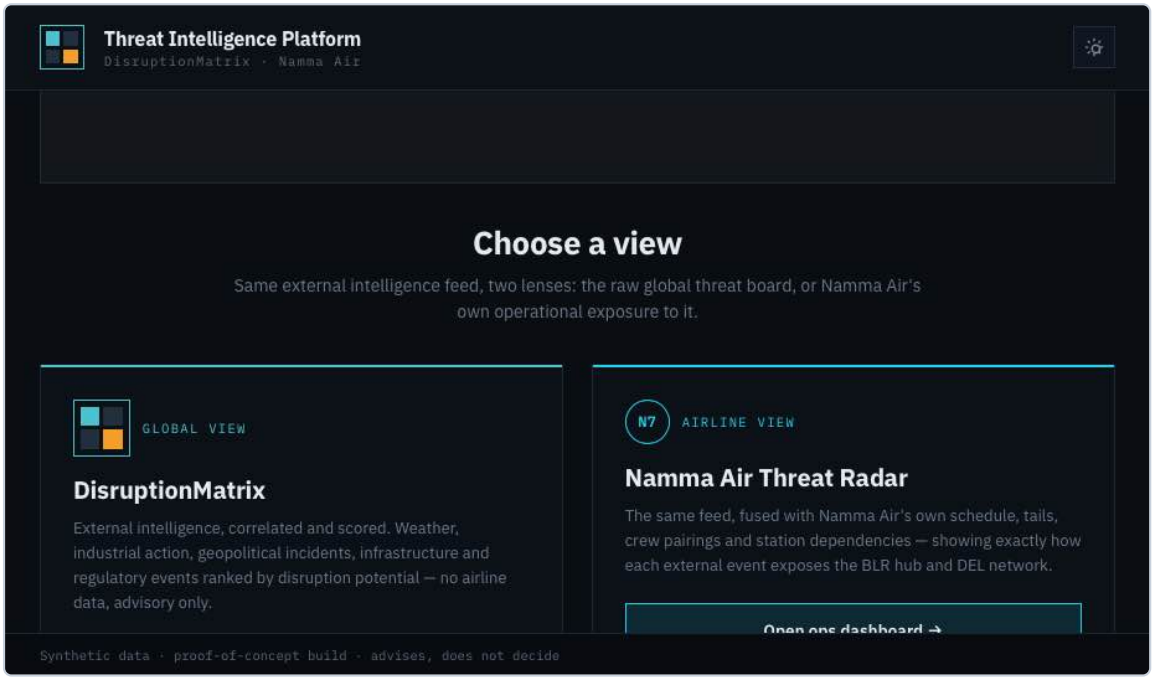


Fig. 1 — Home: two cards (Global view / Airline view), each with a primary action into its dashboard and a secondary link into its map. A droppable hero image sits above the picker.

3. Design system

3.1 Typography

IBM Plex Sans for labels, headings and body copy; **IBM Plex Mono** for every number, score, timestamp, code and status tag — the split is deliberate: mono numerals line up in tables and read as instrument-panel data, distinct from prose.

3.2 Color — DisruptionMatrix (dark, default)

Background #0A0E13 Panel #0C1118 Teal accent #4CC2CE
Critical #F04A52 High #F59E2B Moderate #E0C64A
Low #5B92C9 OK/positive #31C48D

Light mode swaps the surface and text tokens (paper-toned panels, dark ink) but keeps every severity color and the teal accent — only lightened where needed for contrast (teal → `#2F8E9C` in text).

3.3 Color — Namma Air (dark, default)

Background `#0B1220` Panel `#101A2C`
Cyan brand accent `#22D3EE` Critical `#F0495A`
Elevated `#F5A623` Nominal `#34D399`

Namma Air's palette is deliberately distinct from DisruptionMatrix's (navy + cyan vs. charcoal + teal) so a user always knows which lens they're in, while both share the same severity logic: red/amber/green (or high/elevated/watch) means the same thing everywhere.

3.4 Recurring components

Component	Where used	Behavior
Score chip	Every list and header	Mono number, border + 10% fill in its severity color
Pill filter	Category / sort / layout controls	Outlined default, teal/cyan fill + border when active
Confidence bars	Threat rows	4 ticks, filled count = confidence level
Domain / severity dots	Board rows, map legend	Small squares shaded by implication strength
Theme toggle	Every page header	Single icon button — sun shown in dark mode, moon in light mode
Tab strip	Event workspace (DisruptionMatrix)	Underline indicator in teal on the active tab

4. Page-by-page walkthrough

4.1 DisruptionMatrix — Threat Board

Layout: fixed sidebar (board link + live list of active events, sorted by score) · header with clock and "events at VP threshold" indicator · content pane with category/score/sort/layout filters above either a **ranked list** or an **escalation-lanes** Kanban (one column per escalation level).

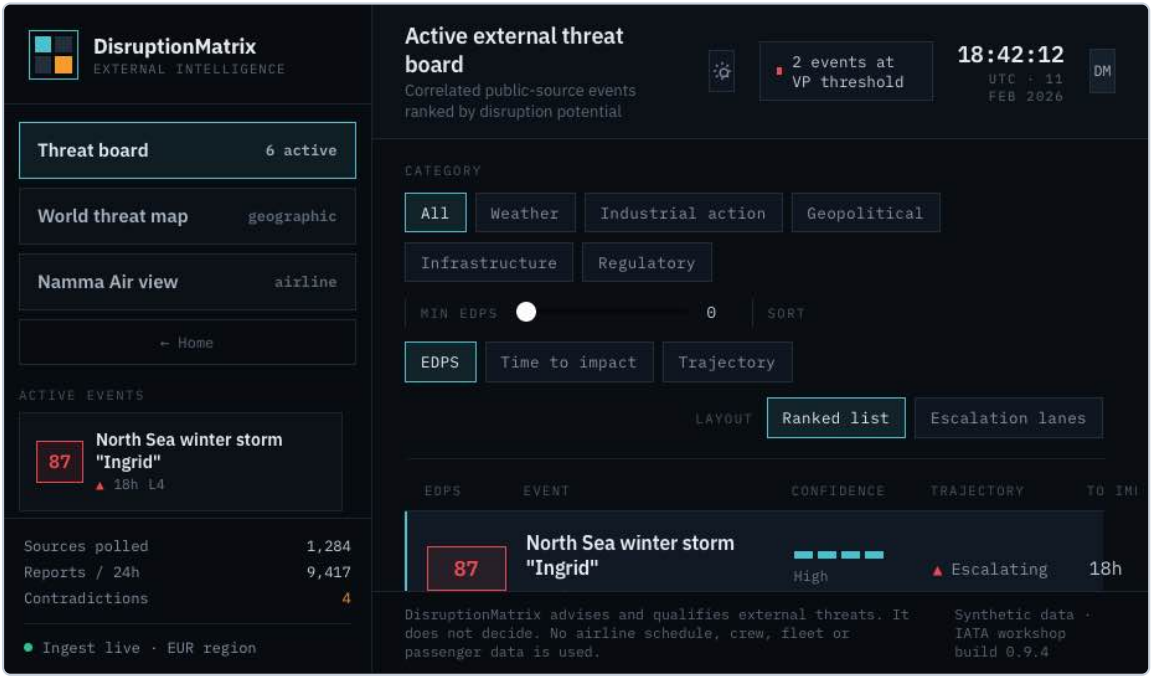


Fig. 2 — Threat board, ranked-list layout. EDPS chip · confidence bars · trajectory arrow · time to impact · domain dots · escalation tag, left to right.

Selecting an event opens its workspace: a persistent header (score, ref, category, summary, four head-stats) above four tabs —

- **Assessment** — score composition (weighted factors, hover for evidence), affected operational domains, airports in the area of effect, airlines operating there, knock-on effect pathway, evidence trail with assumptions/gaps.
- **Escalation & response** — the four-level escalation model (Watch → Assess → Engage → Decide) and a six-phase response process with per-step checkboxes.
- **Event-type SOP** — the airline-configured playbook for that category of event: gates and a time-phased action list.
- **Retrospective** — "when could we have known?": a two-column timeline of intelligence availability vs. organisational action, for post-event review.

4.2 Global Threat Map

Plain Leaflet map (real OpenStreetMap tiles, not a DC) so it can pan/zoom freely. Each event is a marker sized/colored by EDPS with a dashed area-of-effect circle at its real radius; markers auto-declutter into free screen slots with leader lines when close together. Filter by category, toggle station dots, and switch extent (affected region / world / zoom-to-selected). Selecting a marker fills the right-hand card: score, disruption level, confidence, trajectory, impact icons, and an "Open threat detail →" button that deep-links into the board.

4.3 Namma Air Threat Radar

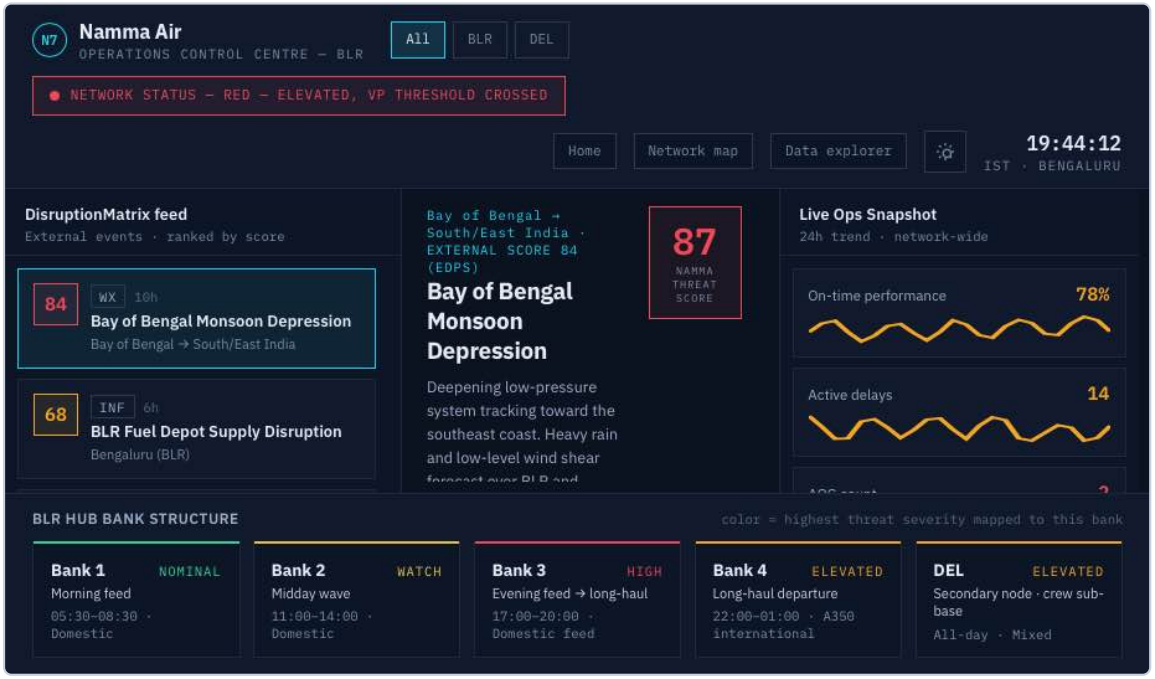


Fig. 3 — Ops dashboard: DisruptionMatrix feed (left) · exposure board for the selected event (center) · live ops snapshot with sparklines (right) · four-bank BLR structure + DEL node (bottom).

The center exposure board is the product's core idea: for the selected external event, it shows the airline-specific **Namma Threat Score** beside the raw EDPS, then five exposure tiers — schedule/fleet, tail/rotation chains, crew pairings near duty limits, connecting pax/cargo, station dependency & escalation — followed by the affected rotations, at-risk crew pairings, and the network's four named structural vulnerabilities (Bank 3→4 fragility, split CAT IIIB certification, single-supplier fuel dependency, LHR overflight exposure). A "Fix schedule →" button on the affected-rotations panel jumps into the Gantt solver, pre-scoped to that event.

4.4 Namma Air Network Map

Leaflet map centered on the BLR/DEL network. Selecting a threat (chip row) draws straight route lines from BLR or DEL to every destination it affects, colored by severity, with a domestic/full-network extent toggle and a hub filter. The side card repeats the exposure summary and affected-flight list from the Radar.

4.5 Namma Air Data Explorer

A tabbed raw-data browser — six datasets (flight schedule, fleet, crew roster, crew pairings, bank structure, threat feed), each a filterable table — plus a seventh, custom view:

Schedule Gantt. Aircraft-by-tail timeline (00:00–24:00); bars are colored by the severity of whatever threat touches that flight, grey if unaffected. A **Run solver** button computes a deterministic recovery action per threat type (delay past a weather window, swap an AOG or uncertified tail, hold

for CAT IIIB availability, absorb an ATFM delay, reroute for added block time, or simply monitor) and renders a second "Proposed schedule" Gantt beside the current one, plus a Changes table (tail and departure time, before → after, action taken). Everything is labeled *proposed, not yet approved* — consistent with the platform's advisory framing throughout.

5. Interaction & state conventions

Pattern	Convention
Selecting an item in a list	Left border + tinted background on the selected row/card; never a modal
Filters	Always additive pills or a single dropdown, never more than one control style per screen
Cross-product navigation	Explicit named links in the header, never hidden in a menu
Theme	One shared key across all six pages; changing it anywhere changes it everywhere on next load
Deep links	Plain query params (`?event=`, `?tab=`, `?threat=`) read on mount, no routing library
Proposed vs. approved	Anything computed by a "solver" or model is visually distinct (dashed/cyan-bordered) and labeled as a proposal

6. Data boundary

DisruptionMatrix holds no airline schedule, crew, fleet or passenger data — stated in its footer on every screen. Namma Air's products are the only place airline operational data appears, and every score they produce is explicitly advisory: the platform surfaces exposure and proposes recovery actions; the OCC decides.